Multiple Target Tracking Based Separation of Micro-Doppler Signals from Coning Target

Shao Changyu, Du Lan, Han Xun and Liu Hongwei
National Lab of Radar Signal Processing
Xidian University
Xi'an, China
Email: dulan@mail.xidian.edu.cn

***Abstract*—Micro-Doppler (m-D) signature based target classification methods have received the intensive attention from the radar automatic target recognition (RATR) community. Since the m-D signals of a coning target with micro-motion are always multicomponent, it is important to separate these m-D frequency components for feature extraction and parameter estimation. This paper develops a separation algorithm based on multiple target tracking (MTT) technology, where the m-D frequency curves are assumed to be the tracks of maneuvering targets. The proposed method shows the good performance on the electromagnetic computation data in the experiments.**

Keywords—micro-motion; micrio-Doppler signals separation; multiple target tracking

I. INTRODUCTION

The spectrum shift of radar echoes in frequency domain is known as Doppler effect, which is usually induced by the translation of a target. When a target or any structure on the target has micro-motion, vibration or rotation, it can induce an additional frequency modulate on the radar echoes. This phenomenon is called micro-Doppler (m-D) effect [1]. M-D effect was first observed in coherent laser radar systems [2]. [3] studied the mathematic principle of the m-D signature induced by a point target with vibration or rotation. M-D signature is now regarded as a proper signature of the target with micro-motion.

Time-frequency (TF) analysis is a primary tool in the analysis of m-D effect. The micro-motion of an equivalent scatter center can be represented as a TF curve in the TF plane. Usually, there are several TF curves when the radar target has more than one equivalent scatter on it, and these TF curves are always overlapped in TF plane for a target with micro-motion. To estimate the target geometry paramter or micro-motion parameter and extract features for target classification, these overlapped TF curves should be separated first. The Hough transformer is used to separate the m-D signals and estimate the micro-motion parameters in [4]. However, this method needs to develop the parameter equation of the micro-motion, which is not easy especially when the micro-motion is complicated. The Viterbi algorithm is applied in [5] to separate the TF curves. This method works well when the TF plane is sharp enough, but may get the incorrect results in the overlapped area of the TF curves or under the low signal-to-noise ratio (SNR) condition, since its tracking results only depend on the locations and the magnitudes of the m-D frequencies.

Based on the equivalent scatter center model, this paper first analyzes the relationship between the m-D frequencies and the motion parameters of a coning target with precession or nutation. Due to the complicated forms of the m-D signals, we proposed a separation algorithm to extract the m-D frequency curves based on the multiple target tracking (MTT) technology. The advantage of this algorithm is that simple motion model is utilized instead of the parameter equations of the target with micro-motion. The performance of the proposed method is evaluated via the electromagnetic computation data in the experiment section.

II. SIGANAL MODELS

As shown in Fig. 1, the origin o of the target coordinate system (x, y, z) is located at the bottom center of the cone. Assume that the azimuth and elevation angle of the radar are α and β, respectively. When the cone does precession, and the spin axis and coning axis are oA and oM respectively. When the cone does nutation, the precession angle θ varies periodically while the cone is doing precession. For the purpose of mathematical analysis, a reference coordinate system (x', y', z') is introduced, which shares the same origin o with the target coordinates. The z' axis and the axis of the cone are superposed together, and the x' axis is perpendicular to Noz' plane, where No is the radar line of sight (LOS).

There are only three equivalent scatter centers of a smooth cone: the conic node A and two intersections of the plane NoA and the bottom edge of the cone (B and C). Usually, the scatter C is occlusive. For convenience, we assume $\alpha = 0°$, and the direction of the radar LOS is $\mathbf{n} = [0 \quad \sin\beta \quad -\cos\beta]^{\mathrm{T}}$.

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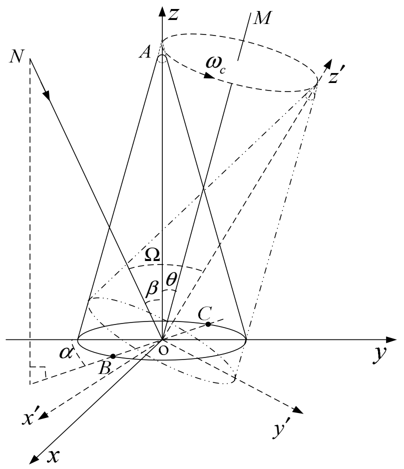

Figure 1. Micro-motion model of a coning target

A. Micromotion model

The cone's nutation is a composition motion of three motions: spinning, coning and oscillating. At the initial moment, the direction of oA is $\mathbf{p}_0 = [0 \quad 0 \quad 1]^{\mathrm{T}}$. At time t, the unit vector becomes [6]

$$\mathbf{p}_{\mathrm{nut}}(t) = \mathbf{R}_{\mathrm{v}}(t)\mathbf{R}_{\mathrm{c}}(t)\mathbf{R}_{\mathrm{s}}(t)\mathbf{p}_0 \tag{1}$$

where, $\mathbf{R}_{\mathrm{s}}(t)$ is the spin motion matrix, $\mathbf{R}_{\mathrm{c}}(t)$ denotes the coning motion matrix, and $\mathbf{R}_{\mathrm{v}}(t)$ is the oscillating motion matrix. If $\mathbf{R}_{\mathrm{v}}(t)$ is an identity matrix, the cone does precession.

B. Micro-Doppler frequency analysis

Assuming that the angle of $\mathbf{p}_{nut}(t)$ and $\mathbf{n}$ is less than $\pi/2$, the direction of radar LOS in the reference coordinates is

$$\mathbf{n}' = \left[0 \quad \left(1-\Omega^2(t)\right)^{1/2} \quad -\Omega(t)\right]^{\mathrm{T}} \tag{2}$$

where,

$$\Omega(t) = \cos\left(\mathbf{p}_{nut}(t) \hat{,} \mathbf{n}\right) = \mathbf{p}_{nut}^{\mathrm{T}}(t)\mathbf{n} \tag{3}$$

At the initial moment, a scatter P on the cone is located at $\mathbf{r}_0' = [x_0 \quad y_0 \quad z_0]^{\mathrm{T}}$ in the reference coordinates. Then the m-D frequency at time t can be calculated as

$$f_{\mathrm{m\text{-}D}}(t) = \frac{2}{\lambda}\frac{dr(t)}{dt} \tag{4}$$

where, $r(t) = \mathbf{n}'^{\mathrm{T}}\mathbf{r}_0'$. Equations (1) ~ (4) show that the m-D signal of the scatter P varies in a complex form rather than a sine. Furthermore, the spinning of the cone does not affect the m-D signals since $\mathbf{R}_s(t)\mathbf{p}_0 = \mathbf{p}_0$.

III. Separation Algorithm

In this section, a novel method to separate the m-D signals is presented. Since the m-D signals are time-varying, short time Fourier transform (STFT) analysis is applied to obtain the TF plane of the echoes. The m-D signals can be represented as TF curves in TF plane. However, the TF curves in TF plane are always overlapped. In order to separate these curves, MTT technology is applied to track them. The m-D frequency points at each time slice in TF plane are seen as 'radar plots' and the TF curves are assumed to be the tracks of targets.

A. MTT technology

Before applying MTT algorithm, constant false alarm rate (CFAR) technology is adopted first on the TF plane to obtain the 'plots'. Since the precision of TF plane is not high enough, it is also necessary to implement plots centroid technology to evaluate the locations of the m-D frequencies.

MTT algorithm is usually used to track multiple targets in radar signal processing. It is a composition algorithm of track initiation, data association, filter algorithm and track elimination. If occlusion is ignored and SNR is high enough, the TF curves in TF plane are always consecutive. So, we do not consider the track initiation and track elimination in this paper.

1) Filter algorithm and motion model: In this paper we adopt the Kalman filter and constant velocity (CV) model. More details about Kalman filter can be found in [7]. As the overlapped area in TF plane is usually dim, many false plots are generated after CFAR. In this situation, constant accelerate model or other high order models are not adopted here because the data association algorithm easily gets the incorrect result based on these models. CV model is shown as (5).

$$\begin{bmatrix} \dot{x} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ \dot{x} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} w(t) \tag{5}$$

where, $w(t) \sim \mathrm{N}(0, \sigma^2)$ is the noise, and x, $\dot{x}$ and $\ddot{x}$ denote the position, velocity and accelerate of the m-D frequency in TF plane, respectively.

2) Data association algorithm: After CFAR and plots centroid technology, it is necessary to automatically associate these raw plots with each m-D signal. This problem is solved using a track association scheme based on the Kalman filter. Here, we use the nearest neighbour data association (NNDA) algorithm. This algorithm need to set a tracking gate which ensures that the correct plots should be received in certain probability. The candidate plots which fall in the tracking gate make (6) hold.

$$g(k) = [\mathbf{y}_k - \hat{\mathbf{y}}_{k/k-1}]^{\mathrm{T}} \mathbf{S}_k^{-1} [\mathbf{y}_k - \hat{\mathbf{y}}_{k/k-1}] \le \varepsilon \tag{6}$$

where, $\varepsilon > 0$ denotes the tracking gate, $\mathbf{y}_k$ is the measurement on the time step k, $\hat{\mathbf{y}}_{k/k-1}$ is the innovation and $\mathbf{S}_k$ is the measurement prediction covariance. Both $\hat{\mathbf{y}}_{k/k-1}$ and $\mathbf{S}_k$ can be calculated in Kalman filter. If only one plot falls into the gate, then the plot is used to update the Kalman filter. If there are more than one candidate plot, the one which minimizes $g(k)$ is used to update the Kalman filter.

B. The proposed separation algorithm

The detailed proposed algorithm is presented as follows:

Step 1 Initializing parameters: set the window length m and the overlap number n of the STFT, initialize the filter parameter and obtain the initial plots using CFAR and plots centroid algorithm. $k = 1$.

Step 2 Appling Fourier Transformer: if $k = m$, do Fourier Transformer, else $k = k + 1$ and repeat this step until $k = m$.

Step 3 CFAR and plots centroid: applying CFAR and plots centroid algorithm on the result of Step2.

Step 4 Data association: NNDA algorithm is utilized to associate the plots to each m-D signals.

Step 5 Kalman filter: update the Kalman filter with the plots associated. Kalman smoother also can be used to get smoother TF curves. The next state of the Kalman filter is predicted.

Step 6 $k = n$, and go to step 2.

IV. EXPERIMENTS AND RESULTS

A. Separation of the m-D signals induced by precession

Assuming that the cone precesses and the coning frequency is 4 Hz. Fig. 2a shows the TF plane of the echoes from the cone target and without noise. It is obvious that the two TF curves in TF plane are overlapped. The result of CFAR and plot centroid is shown in Fig. 2b. Fig. 2c gives the TF curves after data association. In Fig. 2d, the curves are smoothed using Kalman smoother. The two components of the m-D signals in Fig. 2a are separated successfully.

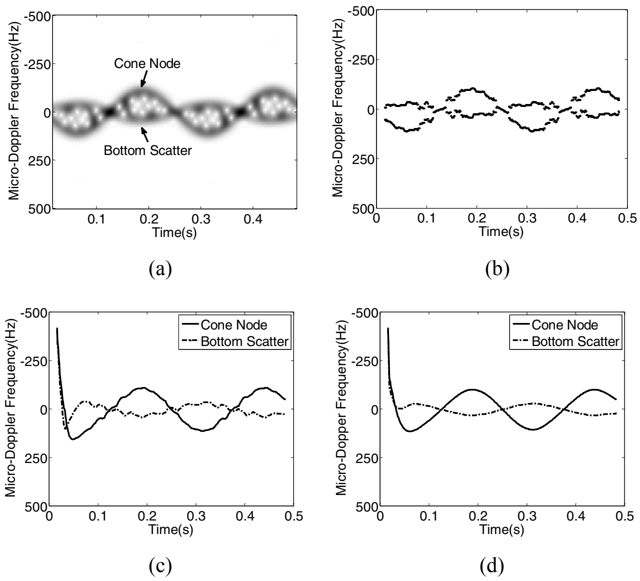

Figure 2. Various figures of experiment A. (a) the TF plane of the echoes from a precession cone target and without noise, (b) the result of CAFR and plot centroid, (c) the result of data association, (d) the result of Kalman smoother.

B. Separation of the m-D signals induced by nutation

Assuming that the cone does nutation, the coning frequency is 4 Hz and the oscillating frequency is 2 Hz. Fig. 3a shows the TF plane of the echoes from the cone target and without noise. Fig. 3b is the result of separation algorithm. Fig. 3b shows that the two TF curves in Fig. 3a are separated correctly. To measure the algorithm performance under noise, we add 5 dB Gauss white noise into the echoes and Fig. 3c shows the result of CFAR and plot centroid. There are many false plots appear in Fig.3c since the SNR is low. Fig. 3d gives the result of the separation algorithm under $\text{SNR} = 5\text{dB}$.

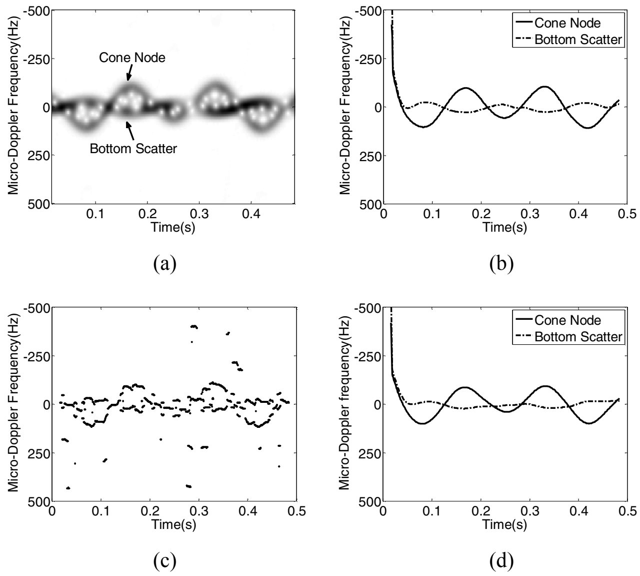

Figure 3. Various figures of experiment B. (a) the TF plane of the echoes from a nutation cone target and without noise, (b) the result of separation algorithm for (a), (c) the result of CAFR and plot centroid under $\text{SNR} = 5\text{dB}$, (d) the result of separation algorithm under $\text{SNR} = 5\text{dB}$.

The results of the experiment A and B show that the proposed separation algorithm can correctly separate the m-D signals whether the coning target does precession or nutation. When the coning target does nutation, the TF curves vary much more complex. So the results are a litter deflected which is mainly induced by Kalman smoother. When the SNR is 5dB , many false plots appear after CFAR and plot centroid. The MTT algorithm can eliminate the false plots availably and separate the m-D signals correctly.

V. CONCLUSION

This paper analyzes the m-D signals of a coning target with precession or nutation based on the equivalent scatter center model. The TF curves in TF plane are not simple sine curves and always overlapped. A separation algorithm based on MTT technology is proposed in this paper to extract the m-D signals. The experimental results show that the proposed algorithm can separate the m-D signals correctly even under $\text{SNR} = 5\text{dB}$. Then our future work will focus on the further feature extraction or parameter estimation based on the separated m-D signals of a coning target.

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